

Exercise 2.9. Prove that if m and n are positive real numbers and $m < n$, then $m^2 < n^2$. You may use the fact that if $a < b$ and c is positive, then $ac < bc$.

Solution.

We are given that m and n are positive real numbers and

$$m < n.$$

We want to prove that

$$m^2 < n^2.$$

Since m is positive, we can multiply both sides of

$$m < n$$

by the positive number m . This gives

$$m^2 < mn.$$

Since n is positive, we can multiply $m < n$ by the positive number n . Thus,

$$mn < n^2.$$

Combining the two inequalities,

$$m^2 < mn < n^2.$$

Therefore,

$$\boxed{m^2 < n^2}.$$

The key reason the proof works is that multiplying an inequality by a positive number preserves the direction of the inequality.